Mathematical overview

A complete matching (bijection) uses every kid exactly once. An order-preserving matching additionally preserves before/after relations. Labels identify kids, not required partners; equal labels in two class lines impose no matching constraint. For the endless class line $1, 2, 3, \ldots$, prepending a star admits the matching star$\mapsto 1$, $n \mapsto n+1$. Thus the order type $1+\omega$ is ω. Appending a star after *every* numbered kid produces a last element; the original class line has none. Hence $\omega+1$ is a different order type.

Two consecutive endless blocks have type $\omega+\omega$. The first kid in the second block has infinitely many predecessors and no immediate predecessor. An endless row of pairs has type $2\cdot\omega=\omega$; every kid has finitely many predecessors. They cannot be matched preserving order. A star in front of any endless row is absorbed in the same way, even a star between two rows: row, star, row has type $\omega+1+\omega=\omega+\omega$, the type of two rows (Problem 4). All these class lines have the same countable cardinal size, using explicit matches that need not preserve order.

Children should discover a structural obstruction or a rule covering every kid. Finite drawn matches are experiments; a uniform assignment and inverse establish an infinite matching. Grades 3–5, with facility interpreting a finite schematic for an endless rule; K–1 has an adult-led route on Problem 1 (see the next page). Ordinal addition here means concatenation. Surreal addition is a distinct commutative operation, so do not transfer these addition identities to Week 83's numbers.

Preparation, launch and readiness

Per pair: numbered kid cards (student page 5), stars and after-every-kid panels (page 6), blank paper for class lines and yarn or dry-erase arrows. For endless rows, draw a few numbered kids followed by an arrow marked "numbers continue forever," as on student pages 2–3; use two colors when needed. A continuation symbol is an instruction, never another kid. A star after every number uses a *separate panel* so children do not read it as after just the printed prefix. Colors and labels identify kids; they do not restrict which kids may be partners. Problem 4's small boxes are sketch space: arrange the cards and hand-drawn endless rows on the table, then sketch the result in those boxes. Provide sufficient table space for two boards together; enlarging to two joined Letter sheets is possible. Physical fit untested.

Prerequisites: compare positions, follow a matching rule, understand "each kid has exactly one partner." Reading is brief and can be read aloud. Reasoning load lies in extending a rule to an arbitrary named position, not counting a printed finite list. One anchored adult can coach pairs; at the upper triplet rotate constructor/checker/questioner.

Launch together: let children reorder a few finite cards and join partners with yarn, then compare the three printed examples: five-to-five in order; five-to-five with the first two partners exchanged; and five-to-six, where one kid must be left out. The last picture shows one failed attempt, and counting proves every attempt fails. Ask the partner to check both unused kids and crossing order. Introduce the continuation-arrow convention *before* asking about an endless class line. Then show what a rule for every kid looks like, on a pair that is not a task: draw two plain endless rows and join top 1 with bottom 2, top 2 with bottom 1, top 3 with bottom 4, and so on; ask, "Who is top kid 20's partner?" (Bottom 19: odd kids go one place right, even kids one place left.) This uses neither the shift nor either obstruction. That it helps is a prediction.

First route after movement: ten minutes on finite Problem 1, 15 minutes prepending and matching, 15 minutes appending and explaining its obstruction. Problem 3's two blocks versus pairs is a second visit or encore; Problem 4 is open construction when the basic convention is controlled. Ordinal names may be introduced after the need arises. Do not require all stations or notation in one hour. Whole-group launch, then selected participating tables only.

K–1, adult-led on Problem 1 (page 1, with the R1–R3 and B1–B3 cards from page 5): lay the three red cards in one line and the three blue cards in another, and let the child join each red kid to one blue kid with yarn. Ask, "How many different ways can you find?" and "Which way has no crossing yarn?" Nothing needs reading. In plain words: there are six ways, because R1 has three choices, then R2 two, then R3 one; only straight across (R1–B1, R2–B2, R3–B3) has no crossings, and that is the way that keeps order. That K–1 children manage this is a prediction.

Facilitator explanations and hints

Problem 1: list the B partners of R1, R2, R3. The six lists are

$$(1,2,3),(1,3,2),(2,1,3),(2,3,1),(3,1,2),(3,2,1).$$

Only the first keeps every pair in order. Exhaustiveness follows from three choices, then two, then one.

Problem 2, prepending: the assignment is onto because kid 1 receives the star and kid $m \geq 2$ receives original $m-1$. It is one-to-one and preserves order. If both lines show numbered kids 1–6, B also shows its star, so matching the drawn cards leaves an apparent unmatched card at the right edge; the repeating rule, not this artificial drawing boundary, resolves the infinite case.

Appending: suppose there were an order-preserving bijection to the ordinary endless class line. The image of the appended last kid would have to be last, but every ordinary numbered kid has a successor. Contradiction. The issue is not inability to physically walk past infinitely many kids.

Ignoring order, star$\mapsto 1$, $n \mapsto n+1$ still gives a size match for the appended-star class line. It moves the last kid to the front, showing why the preservation condition matters. It does not make the class lines have equal order type.

Problem 3: D is drawn on one horizontal line, with all R kids followed by all B kids. The gap separates entire endless blocks, not just the three printed red cards; E interleaves R and B. Matching each named R_n and B_n in D to its namesake in E gives an unrestricted size match. Their positions in E are $2n-1$ and $2n$. This interleaves blocks and violates their original order. No order-preserving match can send B_1 to a kid with only finitely many predecessors, since B_1 has all R kids before it.

Problem 4: write W for an endless row and S for a star. Two rows and up to two stars make ten arrangements of three shapes, and the one-row arrangements of Problem 2 make three more:

Shape	Arrangements
ω (one row)	W, SW, SSW
$\omega+1$	WS, SWS
$\omega+2$	WSS
$\omega\cdot 2$ (two rows)	WW, SWW, WSW, SSWW, SWSW, WSSW
$\omega\cdot 2+1$	WWS, SWWS, WSWS
$\omega\cdot 2+2$	WWSS

Two arrangements of the same shape can be matched keeping order (18 two-row pairs); the other 27 two-row pairs only if order may change. Stars waiting in front of a row join that row; stars after the last row are last kids. The surprise is WSW with WW: the star after the first row comes before every kid of the second, so it joins that row as in Problem 2. Match the first rows kid for kid, then ★ $\mapsto$ B1 and B$n \mapsto$ B$(n+1)$. WW with WWS has a complete match but none that keeps order, because WWS has a last kid and WW does not. Ignoring order, any two arrangements match: stars first, then the rows' kids alternately. WSWS beside WWS needs three "This kid" panels, which page 6 now has.

Hints: point to a last kid if one exists; ask about B1, the first blue kid in D; ask how many kids are before a *named* position. Avoid directing every match in advance. The representation is conceptual; drawn endless rows need not physically contain every kid.

Encore, sources and limits

Encore: attach two, then three kids before or after the endless class line. Compare removing the first kid with removing an appended last kid. Later arrange endlessly many endless blocks as rows with explicit row order; this opens a doorway toward ω^2 but is not required for the present proof.

Sources: Joel David Hamkins, "Counting to Infinity and Beyond" (September 2, 2021), https://jdh.hamkins.org/counting-to-infinity-poster/, for ordinal visualization and continuation to ω^2. Our matching boards and explanatory sequence are independently authored; no poster or video frame is republished. Basic order-type arguments are given fully in this guide.

Pedagogy: *Math Circle by the Bay*, preface p. viii, connects accessible topics to deep mathematics; pp. ix–x discuss materials and varying depth. The grade gate here is our unpiloted design judgment. Existing Week 68 endless graphs explores ends, not these class line order types.

Revised October 23, 2026 from the review card, after the organizer's October 17 approval: "complete" in Problems 1–3, two rows in Problem 4, a third panel on page 6, wider answer lines. Unpiloted. Print US Letter at Actual Size. Physical board fit, yarn handling, table-space requirement, timing and classroom procedure have not been rehearsed. Record whether children interpreted continuation as a rule, whether they distinguished size/order, and any useful invented class lines. A decorative infinity symbol alone is not evidence that the child controlled the investigation.